ZEENATH ARA SYED

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SUMMARY

Robotics software engineer with 7+ years production systems experience including customer-facing deployment projects, completing MSc in Robotics and Embedded AI (Aug 2026). Technical background spans Unity/ROS2 simulation, computer vision, real-time sensor integration, and LLM systems architecture. Proven ability to bridge technical implementation and customer requirements through international deployment experience (Shell Amsterdam), end-to-end project delivery including Site Acceptance Testing, and cross-functional collaboration. Combines deep technical capabilities with customer communication skills and adaptability to deployment environments.

TECHNICAL SKILLS

- **Robotics Systems:** ROS2, MuJoCo, Gazebo, URDF, inverse kinematics, trajectory planning, motion control, sensor fusion
- S**imulation & Visualization:** Unity 3D, digital twin architecture, real-time synchronization, SolidWorks (mechanical CAD)
- C**omputer Vision & AI:** MediaPipe, OpenCV, YOLO, pose estimation, object detection, LLM integration (GPT-4, Llama)
- H**ardware Integration:** IMU sensors (MPU-6050), Arduino, ESP32, camera systems, servo control, tendon-driven mechanisms
- S**oftware Engineering:** Python, JavaScript, C++, SQL, MongoDB, InfluxDB, real-time data pipelines, system architecture

ROBOTICS PROJECTS

Medical Robotics Training Platform - St. James Hospital (Industrial Partnership)

- Architecting robotic eye movement simulator integrating mechanical design (SolidWorks), IMU sensing (MPU-6050), and Unity 3D digital twin with real-time synchronization (<100ms latency, ±2° accuracy) for clinical training.
- Designed tendon-driven actuation system with servo control for multi-axis nystagmus simulation, demonstrating hardware-software integration for medical robotics applications.

Autonomous Surgical Tool Handover System

- Developed proactive robotic assistant using Franka Panda arm in MuJoCo achieving 93% intent recognition accuracy and 127ms response latency through integration of MediaPipe hand tracking, Bayesian inference, and minimum-jerk trajectory prediction.
- Implemented damped inverse kinematics and coordinated motion planning achieving 0.28s temporal synchronization, demonstrating real-time control and human-robot collaboration capabilities.

Computer Vision Benchmarking for Exercise Robotics

- **E**valuated MediaPipe BlazePose, YOLO26, and OpenPose across five metrics for real-world deployment, building Python-Unity 3D evaluation framework with real-time visualization demonstrating system integration and evaluation methodology.
- Validated MediaPipe for exercise applications (4 FPS sustained, 6.8px jitter on CPU) providing deployment-ready recommendations for robotics perception systems.

Vision-Guided Robotic Manipulation

- Developed ROS2-based robotic arm system for pick-and-place operations using point-cloud object detection, trajectory planning, and Gazebo/RViz simulation demonstrating end-to-end robotics development workflow.

PROFESSIONAL EXPERIENCE

Software Development Engineer - L1, Kongsberg Digital **Apr 2022 - Aug 2025**

- Integrated LLM reasoning engine (GPT-4, Llama) into Digital Twin platform, delivering AI-powered decision support for energy sector clients.
- Developed evaluation dashboard measuring LLM performance across precision, fluency, F1 score, and hallucination metrics for customer validation.
- Led Energy Nomination Dashboard integrating real-time sensor streams with 3D visualization, achieving <100ms latency and 25% engagement increase.

Onsite at Shell's Energy Transition Campus, Amsterdam(Nov 2022 - Nov 2023)

- Deployed Digital Twin solutions onsite at Shell's Energy Transition Campus (Amsterdam, Nov 2022-Nov 2023), collaborating directly with innovation team to prototype and validate energy monitoring systems, increasing user engagement 25%

Software Development Engineer (Trainee → L3 → L2), Kongsberg Digital **July 2018 - Mar 2022**

Progressed through three levels delivering Digital Twin solutions with focus on real-time 3D visualization, system integration, and customer delivery.

- Architected real-time spatial visualization rendering millions of pipeline data points for ConEdison, from requirements gathering through SAT.
- Built Unity 3D visualization pipeline integrating time-series databases (InfluxDB, MongoDB) with real-time rendering for robotics simulation applications.
- Led sub-teams and orchestrated software deployments across multiple client engagements.

EDUCATION

MSc - Robotics and Embedded AI *Maynooth University, 2025 - 2026*

B.E - Electronics & Communication Engineering *Reva Institute of Technology, 2013 - 2017*

CERTIFICATIONS

- Fundamentals of Deep Learning - Nvidia(2026)
- Robotics with Python - Udemy (2025)
- Applied Computer Vision: Object Detection and Recognition - Udemy(2025)
- Professional Scrum Master I - Scrum.org (2024)
- Scrum & SAFe PO/PM - Udemy (2024)

ACHIEVEMENTS

- **MUSE Award - Platinum Level (2026):** Recognized by Maynooth University, the highest recognition for leadership and community engagement.
- **Winner - GenAI Hackathon (2024):** Developed a gas lift optimizer that increased profit margins by 20%.
- **Spotlight Awards (2020, 2022):** Recognized for impactful delivery on client-critical products.
- **Ace Award (2022):** For technical leadership in the team and mentoring new engineers.

VOLUNTEERING & LEADERSHIP

- Women Techmakers Ambassador (2023 - Present)
- President - Toastmasters Club (2021 - 2022)
- Mentor & Interviewer - Kongsberg Digital (2021 - 2025)